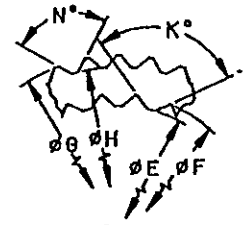
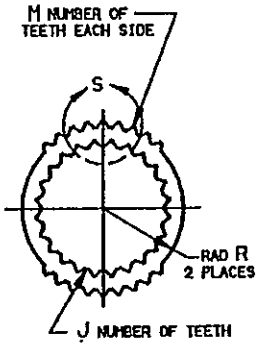
FED. SUP CLASS
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DETAIL S ENLARGED
No specific angularity exists between internal and external serrations. The serrated sectors will be diametrically opposite.



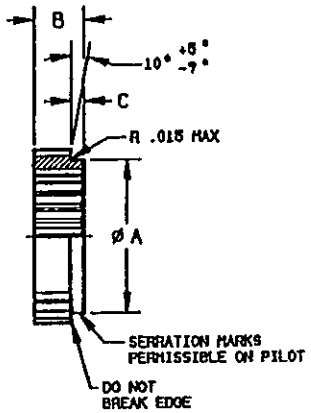


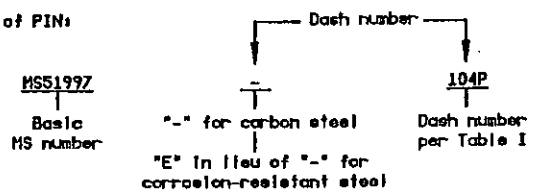
TABLE I. DIMENSIONS

DASH NUMBERS	ØA +.008 -.007	B +.010 -.005	C	ØE +.004 -.003	ØF	ØG +.005 -.004	ØH +.003 -.006	J	K° +1° -2°	M	N°	R RAD REF
102P	.305	.100	.044	.213	.252	.353	.310	13	102°	4	100°	.154
103P	.368	.110	.044	.266	.315	.416	.373	17	86°	5	98°	.186
104P	.430	.138	.044	.323	.364	.478	.435	20	102°	6	97°	.217
105P	.493	.157	.040	.390	.423	.541	.498	24	102°	7	96°	.248
106P	.606	.157	.040	.468	.505	.657	.611	26	102°	7	96°	.304
107P	.731	.161	.040	.574	.613	.782	.736	26	111°	8	95°	.366
108P	.856	.213	.040	.695	.740	.907	.861	30	111°	9	94°	.429
109P	.981	.213	.040	.794	.845	1.034	.986	30	111°	10	96°	.492

REQUIREMENTS:

1. MATERIAL: Steel, carbon, Grade 1117 (UNS G11170) conforming to AMS 5022 or ASTM A108 or Grade 1018 (UNS G10180) conforming to AMS 5069 or ASTM A108.
   Steel, corrosion-resistant, Type A286 (UNS S66286) conforming to AMS 5731, AMS 5734 or AMS 5737, heat treated and aged as applicable.
2. PROTECTIVE COATING OR TREATMENT: Steel, carbon, shall be cadmium plated conforming to QQ-P-416, Type II, Class 3.
   Steel, corrosion-resistant, shall be cleaned, descaled and passivated in accordance with QQ-P-35.
3. SURFACE ROUGHNESS: Machined surfaces to be 125 microinches Ra in accordance with ANSI/ASME B46.1, except serrations.
4. HARDNESS: Carbon steel rings shall be case hardened to 36-45 HRC in accordance with procurement specification.
   Corrosion-resistant steel rings shall have a hardness of 269 HB minimum in accordance with procurement specification.
5. FILLETS: Fillets shall be R .015 maximum.
6. EDGES: Unless otherwise specified, edges broken .003-.015 except serrations.
7. TOLERANCES: Linear dimensions ±.005, angular dimensions ±2°.
8. PART IDENTIFYING NUMBER (PIN): The PIN shall consist of the basic MS number followed by a dash number. Add "E" in lieu of "-" for corrosion-resistant steel.

Example of PIN:



MS51997-104P Ring, Lock, High Strength, Carbon Steel, .478 External Serration Major Diameter.
MS51997E104P Ring, Lock, High Strength, Corrosion-Resistant Steel, .478 External Serration Major Diameter.

© ENTIRE STANDARD REVISED

INCH-POUND

P.A. AR	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
Other Cust AS 99		RING, LOCK, SERRATED-HIGH STRENGTH	MS 51997
PROCUREMENT SPECIFICATION MIL-I-45910	SUPERSEDES:		PAGE 1 OF 2

DD FORM 1 MAY 73 672-1 (COORDINATED) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE 5365-0146

APPROVED 12 DEC 67 REVISED © 19 JAN 89

User activities: Navy - MC

Reviewer activities: Army - AT, AV, CR, ER GL, ME; Air Force - 82; NSA - NS; DLA - IS

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AMSC N/A

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TABLE II
INTERCHANGEABILITY
(see note 4)

DASH NUMBERS	
INACTIVE	NEW
-102	-102P
-103	-103P
-104	-104P
-105	-105P
-106	-106P
-107	-107P
-108	-108P
N/A	-109P
C102	E102P
C103	E103P
C104	E104P
C105	E105P
C106	E106P
C107	E107P
C108	E108P
N/A	E109P

NOTES:
1. Dimensions are in inches.
2. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
3. Referenced Government (or non-Government) documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.
4. The dash numbers "-100" and "C100" series lockrings in the original issue of this standard were inactivated 10 Feb 69. The inactivated lockrings should be used on existing callouts until stock is depleted. Use the new "-100P" and "E100P" series lockrings for replacement of "-100" and "C100" series lockrings in accordance with table II. Use "-100P" and "E100P" series lockrings for all new designs.

APPROVED 12 DEC 67 REVISED © For changes see pages 1 and 2

Reviewer activities:
Army - AT, AV, CR, ER GL, ME
Air Force - 82
NSA - NS
DLA - IS

User activities:
Navy - MC

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

P.A.	AR	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
Other Cust	AS 99		RING, LOCK, SERRATED-HIGH STRENGTH	MS51997
PROCUREMENT SPECIFICATION		SUPERSEDES:		PAGE 2 OF 2
MIL-I-45910				

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